# Signals and Systems — Reference

## how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

| Do | How |
|---|---|
| 1. find the pieces | Each topic opens with a PIECES table. It says where each piece is in your problem, and its color. |
| 2. fill the slots | A piece goes in the slot of the same color: $\omega$ goes in $\boxed{\omega}$. |
| 3. one box at a time | Each box is one move. Do not merge two boxes. |

**Sinusoid colors stay fixed**: amplitude, frequency, phase, sampling rate. Other topics set their colors in their PIECES tables. The number in parentheses in the tables below is the topic page.

## the big picture

Everything in this course is one of two things:

- a **signal**: a quantity that changes over time, like a voltage or a sound.
- a **system**: a machine that takes a signal in and puts a signal out.

The topics build on each other in this order:

1. **How big is a signal?** Energy and power (4–5).
2. **Building blocks.** The button $\delta$, the switch $u$, and the ramp $r$ (10).
3. **Machines.** Systems, block diagrams, difference equations (11).
4. **Describing machines.** Memory, causality, stability, linearity, time invariance (12).
5. **One echo describes the machine.** The impulse response $h$ and the convolution sum $y = x * h$ (13).

The link from 1 to 4: BIBO stability asks if a *bounded* input gives a *bounded* output. That is a question about signal size.

## plain-English glossary

| Symbol | Say it as |
|---|---|
| $\delta[n]$, $\delta(t)$ | **the button**: pressed once at time 0, then released |
| $u[n]$, $u(t)$ | **the switch**: flipped on at time 0 and left on |
| $r[n]$, $r(t)$ | **the ramp**: the running total of the switch, $0, 1, 2, 3, \ldots$ |
| $t$ | the time variable. It sweeps through every time, and is gone after you integrate |
| $T$ | the half-width of a window $[-T, T]$. It is the thing that goes to $\infty$ |
| $t_0$, $n_0$ | "t-naught": one fixed time, such as 3 |
| $N$ | DT window half-width ($2N+1$ samples), or a DT period |
| $\varepsilon$ | "epsilon": a tiny number that we shrink toward 0 |
| $\int \ldots dt$ | add up area as $t$ sweeps (CT) |
| $\sum$ | add up separate terms (DT), the DT twin of $\int$ |
| $dt$ | an infinitely thin slice of time. Not $\Delta t$. |
| $B$ | a ceiling: the input never goes above $B$ in size |
| $h[n]$ | **the echo** (impulse response): what an LTI machine puts out for one button press |
| $x * h$ | "$x$ convolved with $h$: add up one echo per input stem |

## notation: CT and DT side by side

| Item | Continuous time (CT) | Discrete time (DT) |
|---|---|---|
| signal | $x(t)$, round brackets | $x[n]$, square brackets |
| time variable | $t$, any real number | $n$, integers only |
| period | $T$ (seconds) | $N$ (samples, an integer) |
| frequency | $\omega$ (rad/s), $f$ (Hz) | $\hat{\omega}$ (rad/sample) |
| energy, power | integral $\int$ | sum $\sum$ |
| rate of change | derivative $\frac{d}{dt}$ | first difference $x[n] - x[n-1]$ |
| running total | integral $\int_{-\infty}^{t}$ | running sum $\sum_{k=-\infty}^{n}$ |
| button | $\delta(t)$, area 1 | $\delta[n]$, height 1 |

## sinusoid quantities

| Symbol | Name | Unit | Link |
|---|---|---|---|
| $A$ | amplitude | same as $x$ | peak value |
| $\omega$ | angular frequency | rad/s | $\omega = 2\pi f$ |
| $f$ | frequency | Hz = cycles/s | $f = \frac{1}{T}$ |
| $T$ | period | s | $T = \frac{2\pi}{\omega}$ |
| $\phi$ | phase | rad | $\phi = -\omega t_0$ |
| $t_0$ | time shift | s | $t_0 = -\frac{\phi}{\omega}$ |
| $\hat{\omega}$ | DT frequency | rad/sample | $\hat{\omega} = \omega T_s$ |
| $T_s$, $f_s$ | sampling period, rate | s, Hz | $f_s = \frac{1}{T_s}$ |

## trig and complex number facts

| $\theta$ | $0$ | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\pi$ |
|---|---|---|---|---|---|---|
| $\cos\theta$ | $1$ | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$ | $0$ | $-1$ |
| $\sin\theta$ | $0$ | $\frac{1}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | $1$ | $0$ |

$\sin 0 = 0$ and $\cos 0 = 1$. Do not swap them.

| | |
|---|---|
| sine as a cosine | $\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$ |
| negative amplitude | $-A\cos\theta = A\cos(\theta + \pi)$ |
| square of a cosine | $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$ |
| Euler's formula | $e^{j\theta} = \cos\theta + j\sin\theta$ |
| cosine from exponentials | $\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$ |
| sine from exponentials | $\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$ |
| size of $a + jb$ | $\sqrt{a^2 + b^2}$ |
| angle of $a + jb$ | $\arctan\frac{b}{a}$ (check the quadrant) |
| polar form | $C = \lvert C\rvert e^{j\phi}$ |
| size of $e^{j\theta}$ | $1$ |
| whole turns | $e^{j2\pi k} = 1$ for each integer $k$ |

## sums, integrals, limits

| Item | Result |
|---|---|
| infinite geometric sum, $\lvert r\rvert < 1$ | $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ |
| finite geometric sum, $r \neq 1$ | $\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$ |
| count of terms from $-N$ to $N$ | $2N+1$ |
| $\int t^n\, dt$ | $\frac{t^{n+1}}{n+1}$ |
| $\int e^{at}\, dt$ | $\frac{e^{at}}{a}$ |
| $\int_0^{\infty} e^{-at}\, dt, \quad a > 0$ | $\frac{1}{a}$ |
| definite integral | $F(\text{top}) - F(\text{bottom})$ |
| fixed number over a growing $T$ | $\lim_{T\to\infty} \frac{c}{T} = 0$ |
| a growing $T$ alone | $\lim_{T\to\infty} cT = \infty \quad (c > 0)$ |
| $T$ cancels | $\frac{cT}{2T} = \frac{c}{2}$, a constant |

## system property tests

| Property | Ask yourself | Yes / no example |
|---|---|---|
| memoryless | Does $y$ now use only $x$ now? | yes $x^2$ no $x[n-1]$ |
| causal | Does $y$ avoid peeking at the future? | yes $x[n-1]$ no $x[n+1]$ |
| stable | Does a capped input give a capped output? | yes $x^2$ no $\sum_{k\le n} x[k]$ |
| invertible | Can I undo it and get $x$ back? | yes $2x$ no $x^2$ |
| time invariant | Does it ignore the clock? Delay in = delay out | yes $x[n-1]$ no $nx[n]$ |
| linear | Scale in → scale out, add in → add out? | yes $nx[n]$ no $2x+3$ |

## properties of common DT systems

| System | Memoryless | Causal | Linear | Time inv. | Stable |
|---|---|---|---|---|---|
| $x[n-1]$, $x[n]-x[n-1]$, moving average | no | yes | yes | yes | yes |
| accumulator | no | yes | yes | yes | no |
| $nx[n]$ | yes | yes | yes | no | no |
| $x[n]^2$, $2x[n]+3$ | yes | yes | no | yes | yes |
| $x[n+1]$ | no | no | yes | yes | yes |
| $x[2n]$, $x[-n]$ | no | no | yes | no | yes |
| $0.5y[n-1]+x[n]$ (at rest) | no | yes | yes | yes | yes |

## which tool? — given → find

| Given → find | Do this | |
|---|---|---|
| $x(t)$, $T_s \to x[n]$ | put $t = nT_s$ | (1) |
| $A\cos(\omega t+\phi) \to f, T, t_0$ | read, then divide | (2) |
| $f, f_s \to \hat{\omega}$ | $\hat{\omega} = 2\pi f/f_s$ | (3) |
| dies out or finite length | energy $E_\infty$ | (4) |
| periodic or constant | power $P_\infty$ | (4) |
| a graph → energy | square each piece, add | (5) |
| $x(t) \to x(at+b)$ | map each breakpoint | (6) |
| $y = x(at+b) \to x$ | $x(t) = y((t-b)/a)$ | (6) |
| two graphs → $y = Ax(at+b)$ | match two points, solve | (6) |
| signal → period | $2\pi/\omega$, or reduce $\hat{\omega}/2\pi$ | (7) |
| signal → even, odd, neither | compare $x(-t)$ with $x(t)$ | (8) |
| $Ce^{at}$ → real signal | Euler, then the real part | (9) |
| integral with $\delta(t-t_0)$ | the flash: three questions | (10) |
| a pulse graph → formula | a switch on, a switch off | (10) |
| DT stems → buttons | one $x[k]\delta[n-k]$ per stem | (10) |
| graph with corners → formula | a ramp per corner | (10) |
| blocks → one equation | substitute block by block | (11) |
| equation ↔ block diagram | one $D$ per shift | (11) |
| system → properties | ask each question | (12) |
| system → stable? | cap the input with $B$ | (12) |
| equation → $h$ | read coefficients, or run $x = \delta$ | (13) |
| $h$ → causal, stable | $h[n<0] = 0$, $\sum\lvert h\rvert$ | (13) |
| stems $x$, $h \to y$ | table: one row per stem | (13) |
| pulse and formula → $y$ | flip and slide, by regions | (13) |
| two formulas → $y$ | sum $k = 0..n$, geometric | (13) |

The number in parentheses is the topic page.

# 1 — signals and sampling

**THE IDEA**
A **signal** changes over time. **Sampling** takes one snapshot every $T_s$ seconds: the movie $x(t)$ becomes a stack of photos $x[n]$.

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $x(t)$ | the CT signal (the movie) |
| $T_s$ | seconds between photos. Or $T_s = \frac{1}{f_s}$ from the rate |
| $x[n]$ | the DT signal (the photos) |

| CT | DT |
|---|---|
| $x(t)$, round brackets | $x[n]$, square brackets |
| $t$ any real number | $n$ an integer only |

**Trap:** $x[2.5]$ does not exist. Between samples a DT signal is undefined, not zero.
**Trap:** $T_s$ is seconds per photo. $f_s$ is photos per second. Flip one to get the other.

**WORK**

### *Steps — CT → DT (sample)*

**1. READ THE PIECES**
$x(t)$ $T_s$ *(if given $f_s$: $T_s = 1/f_s$)*

↓

**2. SUBSTITUTE**
$x[n] = \boxed{x}\left(n \cdot \boxed{T_s}\right)$ *(write $nT_s$ wherever the formula says $t$)*

↓

**3. SIMPLIFY**
Collect the numbers. The answer has only $n$ in it.

### *Steps — DT → CT (undo)*

**1. READ THE PIECES**
$x[n]$ $T_s$

↓

**2. SOLVE FOR $n$**
photo $n$ was taken at $t = nT_s$, so $n = \dfrac{t}{\boxed{T_s}}$

↓

**3. SUBSTITUTE**
$x(t) = \boxed{x}\left[\dfrac{t}{\boxed{T_s}}\right]$ *(write $t/T_s$ wherever the formula says $n$)*

## 2 — sinusoids

**THE IDEA**
A dot on a spinning wheel. Its height over time is a sinusoid. Wheel size: amplitude. Spin rate: frequency. Where the dot starts: phase.

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $A$ | the number in front of cos. The peak |
| $\omega$ | the number multiplying $t$ (rad/s) |
| $\phi$ | the number added inside, sign included (rad) |

**Exact form first**: $x(t) = A\cos(\omega t + \phi)$.

| If you have | Rewrite as |
|---|---|
| $A\sin\theta$ | $A\cos\left(\theta - \frac{\pi}{2}\right)$ |
| $-A\cos\theta$ | $A\cos(\theta + \pi)$ |

### *Which way?*

| Given → find | Do this |
|---|---|
| $\omega \to f$ | $f = \dfrac{\omega}{2\pi}$ |
| $f \to \omega$ | $\omega = 2\pi f$ |
| $f \to T$ / $T \to \omega$ | $T = \dfrac{1}{f}$ / $\omega = \dfrac{2\pi}{T}$ |
| $\phi \to t_0$ | $t_0 = -\dfrac{\phi}{\omega}$ |
| $t_0 \to \phi$ | $\phi = -\omega t_0$ |

| Quick example | Work → answer |
|---|---|
| $3\cos(4\pi t - \frac{\pi}{3})$ | $A = 3$, $\omega = 4\pi$, $\phi = -\frac{\pi}{3}$ |
| $4\sin(10t)$ | $= 4\cos(10t - \frac{\pi}{2})$: 4, 10, $-\frac{\pi}{2}$ |

**Trap:** Negative $\phi$ means $t_0 > 0$: delayed, moves right.
**Trap:** The first peak is at $t = t_0$. That is the shift.

**WORK**

### *Steps — read the parts*

**1. EXACT FORM**
Rewrite to $\boxed{A}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$. Read the three pieces.

↓

**2. FREQUENCY**
$f = \dfrac{\boxed{\omega}}{2\pi}$ Hz *(one turn is $2\pi$ rad)*

↓

**3. PERIOD**
$T = \dfrac{1}{f} = \dfrac{2\pi}{\boxed{\omega}}$ s

↓

**4. TIME SHIFT**
$t_0 = -\dfrac{\boxed{\phi}}{\boxed{\omega}}$ s *($A\cos(\omega(t - t_0))$)*

### *Steps — build from the parts*

**1. AMPLITUDE**
$A$ = the peak height.

↓

**2. FREQUENCY**
$\boxed{\omega} = \dfrac{2\pi}{T}$ or $2\pi f$.

↓

**3. PHASE**
$t_0$ = time of the first peak. $\boxed{\phi} = -\boxed{\omega}t_0$

↓

**4. ASSEMBLE**
$x(t) = \boxed{A}\cos\left(\boxed{\omega}t + \boxed{\phi}\right)$

# 3 — DT sinusoids, DT frequency, aliasing

**THE IDEA**
Photograph the spinning wheel with a strobe. In the photos you only see how far it turned between photos. That angle per photo is the **DT frequency** $\hat{\omega}$ (rad/sample).

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $\omega$ or $f$ | the CT frequency (rad/s or Hz) |
| $f_s$ or $T_s$ | the sampling rate or period |
| $\hat{\omega}$ | the DT frequency: the number multiplying $n$ |

| Given → find | Do this |
|---|---|
| $f$, $f_s \to \hat{\omega}$ | $\hat{\omega} = \dfrac{2\pi f}{f_s}$ |
| $\omega$, $T_s \to \hat{\omega}$ | $\hat{\omega} = \omega T_s = \dfrac{\omega}{f_s}$ |
| $\hat{\omega}$, $f_s \to f$ | $f = \dfrac{\hat{\omega} f_s}{2\pi}$ (the lowest answer) |

**Aliasing**: $\hat{\omega}$, $\hat{\omega} + 2\pi k$ and $-\hat{\omega}$ give the same photos. In Hz: $\pm f + k f_s$ all look the same.
**Trap:** DT → CT has more than one answer. Any alias fits.
**Trap:** cos is even, so $f_s - f$ aliases to $f$ too.

**WORK**

### *Steps — CT → DT frequency*

**1. READ THE PIECES**
$\omega$ (or $f$) $\quad$ $f_s$

↓

**2. DIVIDE**
$$\boxed{\hat{\omega}} = \frac{\boxed{\omega}}{\boxed{f_s}} = \frac{2\pi \boxed{f}}{\boxed{f_s}}$$

↓

**3. WRITE $x[n]$**
$x[n] = A\cos\left(\boxed{\hat{\omega}} n + \phi\right)$ *($A$ and $\phi$ do not change)*

### *Steps — DT → CT frequency*

**1. READ THE PIECES**
$\hat{\omega}$ $\quad$ $f_s$

↓

**2. MULTIPLY**
$$f = \frac{\boxed{\hat{\omega}} \cdot \boxed{f_s}}{2\pi}$$ *(the lowest answer. Others: $\pm f + k f_s$)*

### *Steps — does $f$ alias to $f'$?*

**1. DT FREQUENCY**
$$\boxed{\hat{\omega}} = \frac{2\pi \boxed{f}}{\boxed{f_s}}$$

↓

**2. REMOVE WHOLE TURNS**
Subtract $2\pi$ until $\hat{\omega}$ is in $(-\pi, \pi]$.

↓

**3. COMPARE**
Same $|\hat{\omega}|$ as $f' \Rightarrow$ same photos (aliases).

# 4 — energy and power

**THE IDEA**
Think of $|x|^2$ as money spent per day, forever. **Energy** is the total. **Power** is the average per day. Spend \$100 once: total finite, average $0$. Spend \$5 a day: total $\infty$, average \$5.

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| on at | where the signal starts. A switch $u(t-a)$ turns on at $a$ |
| off at | where it ends ($\infty$ if never) |
| $\|x\|^2$ | square the signal first |
| $T$ (or $N$) | the window half-width. It goes to $\infty$ |

### *Classify by shape first*

| The signal | Result |
|---|---|
| dies out, or finite length | **energy signal**: $E$ finite, $P = 0$ |
| periodic, or constant | **power signal**: $P$ finite, $E = \infty$ |
| grows without limit ($x = t$) | **neither**: both $\infty$ |

### *Shortcuts*

| Signal | Result |
|---|---|
| sinusoid, amplitude $A$ | $P = \frac{A^2}{2}$ |
| constant $c$ | $P = c^2$ |
| $Ae^{-at}u(t)$, $a > 0$ | $E = \frac{A^2}{2a}$ |
| $a^n u[n]$, $\|a\| < 1$ | $E = \frac{1}{1-a^2}$ |

**Trap:** Square first, then add. Negatives count positive.
**Trap:** "16 forever" is the power. The energy is $\infty$.
**Trap:** A final $E$ or $P$ never contains $T$ or $N$. $\frac{c}{T} \to 0$.
**Trap:** Write the label: energy signal, power signal, or neither.

**WORK**

### *Steps — energy*

**1. CLASSIFY**
Use the shape table. Periodic or constant → go to power.

↓

**2. READ THE PIECES**
on at, off at, and $|x|^2$

↓

**3. INTEGRATE (DT: SUM)**
$$E = \int_{\text{on}}^{\text{off}} |x|^2 \, dt \qquad E = \sum_{\text{on}}^{\text{off}} |x|^2$$
*(where $x = 0$ it adds nothing, so the limits shrink)*

↓

**4. LABEL**
$E$ finite $\Rightarrow P = lim\, \frac{E}{2T} = 0 \Rightarrow$ energy signal

### *Steps — power*

**1. WINDOW TOTAL**
$$\int_{-T}^{T} |x|^2 \, dt \qquad \text{DT: } \sum_{-N}^{N} |x|^2$$

↓

**2. DIVIDE BY THE WINDOW**
CT: $\div\, 2T$ DT: $\div\, (2N+1)$

↓

**3. LIMIT**
Let $T \to \infty$. $T$ must cancel or vanish.

↓

**4. LABEL**
$P$ finite, nonzero $\Rightarrow E = \infty \Rightarrow$ power signal

### *Steps — periodic shortcut*

**1. ONE PERIOD**
$$P = \frac{1}{T_0} \int_{\text{one period}} |x|^2 \, dt$$ *(DT: $\frac{1}{N} \sum$ over one period)*

# 5 — energy from a graph

**THE IDEA**
Do not write one big integral. Chop the graph into straight pieces, find each piece's energy from the shortcut table, and add. In DT there are no widths: each stem's energy is its height squared.

**EXPLANATION**

### *Pieces (one per straight piece)*

| Piece | Where to find it |
|---|---|
| $A$ | the height at the start of the piece |
| $B$ | the height at the end of the piece |
| $D$ | the length (duration) of the piece |
| $h$ | DT: the height of one stem |

### *Shortcut table (the templates)*

| Piece of the graph | Energy of that piece |
|---|---|
| flat at level $A$ | $A^2D$ |
| ramp $0 \to A$ (or $A \to 0$) | $\dfrac{A^2D}{3}$ |
| line $A \to B$ | $\dfrac{D}{3}\left(A^2+AB+B^2\right)$ |
| DT stem of height $h$ | $h^2$ |

Line through $(t_1, x_1)$ with slope $m$: $x = x_1 + m(t - t_1)$, if you want to check by integral.
**Trap:** A squared ramp is a curve: $\div 3$, not $\div 2$.
**Trap:** Negative heights count positive: $(-2)^2 = 4$.
**Trap:** If the graph repeats forever, $E = \infty$. Find the power over one period.

**WORK**

### *Steps — energy from a graph*

**1. CHOP**
Split the graph at every corner and jump. Each piece is flat, a ramp, or a line.

↓

**2. READ EACH PIECE**
$A$, $B$, $D$ for each piece. DT: each $h$.

↓

**3. SHORTCUT EACH PIECE**
flat: $\boxed{A}^2\boxed{D}$ ramp: $\dfrac{\boxed{A}^2\boxed{D}}{3}$ line: $\dfrac{\boxed{D}}{3}\left(\boxed{A}^2+\boxed{A}\boxed{B}+\boxed{B}^2\right)$

↓

**4. ADD**
$E =$ sum of the pieces. DT: $E = \sum \boxed{h}^2$

↓

**5. LABEL**
Finite length ⇒ energy signal.

### *Steps — a repeating graph (power)*

**1. ONE PERIOD**
Find the period $T_0$. Chop one period only.

↓

**2. ENERGY OF ONE PERIOD**
Steps 2–4 above, over one period.

↓

**3. DIVIDE**
$P = \dfrac{E_{\text{one period}}}{T_0}$ ⇒ power signal

# 6 — transformations of the time variable

**THE IDEA**
The shape does not change, only *when* things happen. Ask each corner of the graph one question: **where did this point move to?** It keeps its height and lands at a new time.

## Sketch $x(at+b)$ from $x(t)$

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $t_{\text{old}}$ | each corner or jump of the $x(t)$ graph |
| $a$ | the number multiplying $t$ inside |
| $b$ | the number added inside, sign included |

| Inside | Effect |
|---|---|
| $x(t-t_0)$, $t_0>0$ | moves right (delay) |
| $x(t+t_0)$, $t_0>0$ | moves left (advance) |
| $x(-t)$ | mirror about $t=0$ |
| $x(at)$, $\lvert a\rvert>1$ / $<1$ | compress / stretch |

**Trap:** The sign inside is opposite to the direction: $x(t-3)$ moves right.
**Trap:** Scale-then-shift is wrong. The point map does it in one step.
**Trap:** DT: $x[2n]$ keeps only the even samples. It cannot be undone.

**WORK**

**1. LIST THE CORNERS**
Each corner $t_{\text{old}}$ of $x(t)$, with its height.

↓

**2. READ $a$, $b$**
Match the inside to $\boxed{a}t+\boxed{b}$.

↓

**3. MAP EACH CORNER**
$$t_{\text{new}}=\frac{\boxed{t_{\text{old}}}-\boxed{b}}{\boxed{a}}$$ *(heights do not change)*

↓

**4. DRAW**
Connect the moved corners in the same order. If $a<0$ the order reverses.

↓

**5. DT ONLY**
Keep only integer $t_{\text{new}}$. A non-integer sample is lost.

## Undo, or find $y=Ax(at+b)$ from two graphs

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $A$ | height of $y$ ÷ height of $x$. Upside down: $A<0$ |
| $a$ | negative if $y$ is a mirror of $x$ |
| $(t_1,\tau_1)$, $(t_2,\tau_2)$ | two matching points: $x$ at $t_1$ moved to $\tau_1$ in $y$ |

| Given → find | Do this |
|---|---|
| $x(t)\to x(at+b)$ | map each corner (above) |
| $y(t)=x(at+b)\to x$ | $x(t)=y\left(\frac{t-b}{a}\right)$ |
| two graphs $\to A,a,b$ | two matching points (right) |

**WORK**

### *Steps — undo $y=x(at+b)$*

**1. TEMPLATE**
$$x(t)=y\left(\frac{t-\boxed{b}}{\boxed{a}}\right)$$ *(solve $at+b=s$ for $t$, rename $s\to t$)*

### *Steps — find $A$, $a$, $b$*

**1. HEIGHT**
$$\boxed{A}=\frac{\text{height of } y}{\text{height of } x}$$

↓

**2. MIRROR?**
Mirrored $\Rightarrow \boxed{a}<0$.

↓

**3. TWO EQUATIONS**
$\boxed{a}\boxed{\tau_1}+\boxed{b}=\boxed{t_1}$ $\quad$ $\boxed{a}\boxed{\tau_2}+\boxed{b}=\boxed{t_2}$
*($y$ at $\tau$ is $x$ at $a\tau+b$)*

↓

**4. SOLVE**
Subtract the equations for $a$. Back-substitute for $b$.

↓

**5. ASSEMBLE, CHECK**
$y(t)=\boxed{A}x(\boxed{a}t+\boxed{b})$. Test a third point.

# 7 — periodic signals

**THE IDEA**
A periodic signal is a song on repeat. Slide it by one period and nothing changes. The **fundamental period** is the shortest loop.

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $\omega$ | CT: the number multiplying $t$ |
| $\hat{\omega}$ | DT: the number multiplying $n$ |
| $m/N$ | $\hat{\omega}/2\pi$ as a fraction in lowest terms |

| Test | Means |
|---|---|
| $x(t+T)=x(t)$ | CT periodic with period $T$ |
| $x[n+N]=x[n]$, $N$ an integer | DT periodic |

| Quick example | Work → answer |
|---|---|
| $\cos(\frac{4\pi}{6}n)$ | $\frac{4}{12}=\frac{1}{3} \Rightarrow N=3$ reduce first |
| $\cos(n)$ | $\frac{1}{2\pi}$ not rational: not periodic |

**Trap:** A CT sinusoid is always periodic. A DT sinusoid is not always.
**Trap:** Reduce the fraction before reading $N$.

**WORK**

### *Steps — CT period*

**1. EACH TERM**
$T_i = \dfrac{2\pi}{\boxed{\omega_i}}$

↓

**2. COMBINE**
One term: $T = T_1$. A sum: $T = \text{lcm}(T_1, T_2, \ldots)$ *(the first time all loops restart together)*

### *Steps — DT period*

**1. DIVIDE BY $2\pi$**
$\dfrac{\boxed{\hat{\omega}}}{2\pi}$

↓

**2. RATIONAL?**
Not a fraction of integers $\Rightarrow$ not periodic. Stop.

↓

**3. REDUCE**
Write it as $\dfrac{m}{\boxed{N}}$ in lowest terms.

↓

**4. READ $N$**
The period is the denominator $N$. *($N$ photos hold $m$ whole turns)*

↓

**5. A SUM (DT)**
$N = \text{lcm}(N_1, N_2, \ldots)$

# 8 — even and odd signals

**THE IDEA**
**Even**: a mirror on the vertical axis. **Odd**: spin the graph half a turn about the origin and it lands on itself. Every signal splits into one even part plus one odd part.

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| even pile | terms with cos, $t^2$, $\lvert t\rvert$, constants |
| odd pile | terms with sin, $t$, $t^3$ |
| $x(-t)$ | replace every $t$ with $-t$ |

| Test | Result |
|---|---|
| $x(-t) = x(t)$ | even |
| $x(-t) = -x(t)$ | odd |
| neither holds | neither |

**Products work like signs**: even·even = odd·odd = even. even·odd = odd.
**Trap:** An odd signal is 0 at the origin.
**Trap:** DT: read $x[-n]$ from $x$ at the opposite index.

**WORK**

### *Steps — test*

**1. FLIP**
Write $x(-t)$. Simplify.

↓

**2. COMPARE**
Same as $x(t)$: even. Same with every sign flipped: odd. Else: neither.

### *Steps — split (formula)*

**1. SORT THE TERMS**
Each term goes in the even pile or the odd pile.

↓

**2. READ THE PARTS**
$\mathcal{E}v\{x\} =$ [even pile] $\quad\quad$ $\mathcal{O}d\{x\} =$ [odd pile]

### *Steps — split (any signal, DT table)*

**1. TABLE**
Rows $n$, $x[n]$, $x[-n]$ for each $n$.

↓

**2. EVEN PART**
$\mathcal{E}v = \dfrac{x[n]+x[-n]}{2}$ in each column

↓

**3. ODD PART**
$\mathcal{O}d = \dfrac{x[n]-x[-n]}{2}$ in each column

↓

**4. CHECK**
$\mathcal{E}v + \mathcal{O}d = x[n]$ in each column. $\mathcal{O}d[0] = 0$.

# 9 — complex exponentials

**THE IDEA**
$e^{j\theta}$ is an arrow of length 1 at angle $\theta$. Let $\theta = \omega t$ and it spins. Its shadow on the horizontal axis is a cosine. Multiply by $e^{\sigma t}$ and it also grows or shrinks: a spiral.

**EXPLANATION**

## *Pieces*

| Piece | Where to find it |
|---|---|
| $\lvert C\rvert$ | length of $C$: $\sqrt{a^2+b^2}$ |
| $\phi$ | angle of $C$: arctan $\frac{b}{a}$, check the quadrant |
| $\sigma$ | real part of $a$: growth (+) or decay (−) |
| $\omega$ | imaginary part of $a$: spin rate |

$x(t) = Ce^{at}, \quad C = \lvert C\rvert e^{j\phi}, \quad a = \sigma + j\omega.$

| Case | Real part looks like |
|---|---|
| $\omega = 0$ | real exponential: grows or decays |
| $\sigma = 0$ | sinusoid, $T = \frac{2\pi}{\lvert\omega\rvert}$ |
| both | damped sinusoid, envelope $\lvert C\rvert e^{\sigma t}$ |

**DT**: $x[n] = C\alpha^n$. $\lvert\alpha\rvert < 1$ decays, $> 1$ grows, $\alpha < 0$ flips sign each step. $e^{j(\hat{\omega}+2\pi)n} = e^{j\hat{\omega}n}$.
**Trap:** $e^{j2\pi k} = 1$ for every integer $k$. $\lvert e^{j\theta}\rvert = 1$.

**WORK**

## *Steps — $Ce^{at}$ → real signal*

**1. POLAR $C$**
$C = \boxed{\lvert C\rvert} e^{j\boxed{\phi}}$

↓

**2. SPLIT $a$**
$a = \boxed{\sigma} + j\boxed{\omega}$

↓

**3. REAL PART**
$\mathrm{Re}\{Ce^{at}\} = \boxed{\lvert C\rvert} e^{\boxed{\sigma}t} \cos\left(\boxed{\omega}t + \boxed{\phi}\right)$

## *Steps — real signal → $Ce^{at}$*

**1. MATCH**
Read $\lvert C\rvert$, $\sigma$, $\omega$, $\phi$ off $\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi)$.

↓

**2. ASSEMBLE**
$C = \boxed{\lvert C\rvert} e^{j\boxed{\phi}} \qquad a = \boxed{\sigma} + j\boxed{\omega}$

## *Steps — cosine or sine → two exponentials*

**1. EULER**
$\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2} \qquad \sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j}$

↓

**2. PLACE**
$\boxed{A}\cos(\boxed{\omega}t) = \dfrac{\boxed{A}}{2} e^{j\boxed{\omega}t} + \dfrac{\boxed{A}}{2} e^{-j\boxed{\omega}t}$

# 10 — the button, the switch, and the ramp

**THE IDEA**
$\delta$ is a **button**: pressed once at time $0$. $u$ is a **switch**: flipped on at $0$ and left on. $r$ is a **ramp**: the running total of the switch. Each is the running total of the one before. Everything else is built from these three.

## Where is it? Stems as buttons, pulses as switches

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $h$ | the height of a stem (the coefficient) |
| $k$ | the position: set the inside to $0$ |
| $a$, $b$ | switch on at $a$, off at $b$ |

| Inside | Position, direction |
|---|---|
| $n-3$ | at $n=3$: delay, right |
| $n+2$ | at $n=-2$: advance, left |

| Given → find | Do this |
|---|---|
| $u \to \delta$ (CT) / (DT) | $\frac{du}{dt}$ / $u[n]-u[n-1]$ |
| $\delta \to u$ (CT) / (DT) | $\int_{-\infty}^{t} \delta$ / $\sum_{k \le n} \delta[k]$ |

**Trap:** The off-switch's own position is already off: $u[n]-u[n-4]$ is $0$ at $n=4$.
**Trap:** $u[n]$ is $0,0,1,1,1,\ldots$ It does not climb. The ramp climbs.

**WORK**

### *Steps — stems → buttons*

**1. READ EACH STEM**
height $h$ at position $k$

↓

**2. ONE BUTTON PER STEM**
$x[n] = \sum \boxed{h}\,\delta[n-\boxed{k}]$ *(no stem, no term)*

### *Steps — buttons → stems*

**1. EACH TERM**
coefficient = height $h$ inside $=0 \Rightarrow$ position $k$

### *Steps — pulse from switches*

**1. TEMPLATE**
DT: $u[n-\boxed{a}]-u[n-\boxed{b}]$ is on for $\boxed{a} \le n \le \boxed{b}-1$
CT: $u(t-\boxed{t_1})-u(t-\boxed{t_2})$ is on for $\boxed{t_1} \le t < \boxed{t_2}$

↓

**2. HEIGHT**
Multiply the pair by the pulse height.

## Sifting: the camera flash

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $t_0$ | the flash: set the inside of $\delta$ to $0$ |
| lo, hi | the limits of the integral (the window) |
| $g(t)$ | everything multiplying $\delta$ |

CT button: area $1$ (not height). $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$.

| Quick example | Work → answer |
|---|---|
| $\int_0^5 e^t\,\delta(t+1)\,dt$ | flash $-1$, outside $[0,5]$: $0$ |
| $\sum_n n^2\,\delta[n-3]$ | flash $3$, all $n$: $3^2$ |

**Trap:** The answer has no $\delta$ left in it.

**WORK**

**1. WHERE IS THE FLASH?**
Set the inside of $\delta$ to $0$: $t = \boxed{t_0}$

↓

**2. IN THE WINDOW?**
Is $\boxed{t_0}$ in $[\boxed{\text{lo}},\boxed{\text{hi}}]$? No $\Rightarrow$ the answer is $0$.

↓

**3. WHAT DOES IT CAPTURE?**
$\int \boxed{g(t)}\,\delta(t-\boxed{t_0})\,dt = \boxed{g}(\boxed{t_0})$ *(plug $t_0$ into everything except $\delta$)*

## Graphs from ramps and switches

EXPLANATION

### *Pieces*

| Piece | Where to find it |
|---|---|
| corner | where the slope changes |
| $\Delta$slope | slope after $-$ slope before |
| jump | where the graph jumps, and by how much (up $+$) |

| Form | Formula |
|---|---|
| ramp | $r(t) = t\,u(t), \quad r[n] = n\,u[n]$ |
| CT chain | $\frac{dr}{dt} = u, \quad \frac{du}{dt} = \delta$ |
| DT chain | $r[n] - r[n-1] = u[n-1]$ |
| DT running sum | $r[n] = \sum_{k=-\infty}^{n-1} u[k]$ |

**Trap:** The DT ramp sum stops at $n-1$.
**Trap:** At a peak the slope change is double: $+1 \to -1$ is $-2$.

WORK

### *Steps — graph → formula*

**1. WALK LEFT TO RIGHT**
List every corner and every jump.

↓

**2. SLOPE CHANGE AT EACH CORNER**
$\boxed{\Delta}$ = slope after $-$ slope before

↓

**3. ASSEMBLE**
$x(t) = \sum \boxed{\Delta} r(t - \boxed{\text{corner}}) + \sum \boxed{\text{jump}}\, u(t - \boxed{\text{where}})$

↓

**4. CHECK**
The slope changes sum to $0$ if the graph ends flat at $0$.

### *Steps — formula → graph*

**1. EACH RAMP IS A CORNER**
$\boxed{\Delta} r(t - \boxed{\text{corner}})$: the slope changes by $\Delta$ there.

↓

**2. EACH SWITCH IS A JUMP**
$\boxed{\text{jump}}\, u(t - \boxed{\text{where}})$

↓

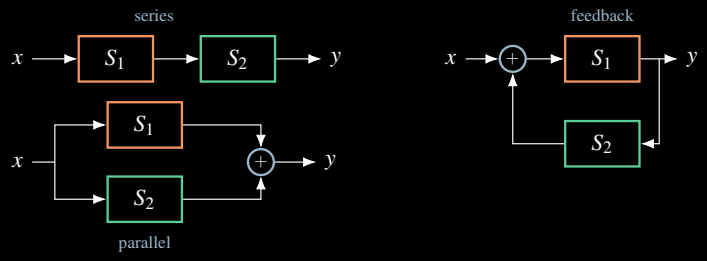
**3. DRAW**
Start flat at $0$. Add each change in time order.

# 11 — systems and interconnections

**THE IDEA**
A **system** is a machine: a signal goes in, another comes out. A **difference equation** is its recipe: how to cook $y[n]$ from inputs, and sometimes from its own past outputs.

## Hookups: series, parallel, feedback

**EXPLANATION**




### *Pieces*

| Piece | Where to find it |
|---|---|
| $S_1$ | the first machine (or top branch) |
| $S_2$ | the second machine (or bottom branch, or the loop) |
| start | a recursive system's start value, $y[-1] = 0$ at rest |

**Trap:** Shifting $w[n] = 2x[n]$ to $w[n-1]$ shifts every $n$: $2x[n-1]$.

**WORK**

### *Steps — series*

**1. NAME THE MIDDLE**
$w =$ output of $S_1$.

**2. SHIFT AS NEEDED**
Write $w[n-1]$, $w[n-2]$, … by replacing every $n$ in $S_1$.

**3. SUBSTITUTE**
Put $w$ into $S_2$'s input slot. The answer has only $x$ and $y$.

### *Steps — parallel*

**1. ADD THE BRANCHES**
$y = \boxed{y_1} + \boxed{y_2}$ *(both get the same $x$)*

### *Steps — run a recursive system*

**1. START AT REST**
$y[-1] = \boxed{0}$ (or the given start).

**2. ONE STEP AT A TIME**
$y[0]$ from $x[0]$ and $y[-1]$. Then $y[1]$ from $x[1]$ and $y[0]$. …

**3. PATTERN**
Spot the rule (often $c\,a^n u[n]$). Check one more $n$.

## Diagram ↔ equation

**EXPLANATION**

### *Pieces*

| Block | Does |
|---|---|
| adder (circle) | sums its inputs |
| gain (box $a$) | $a$·input |
| unit delay (box $D$) | $x[n] \to x[n-1]$ |
| branch dot | copies a signal |

| System | Equation |
|---|---|
| first difference | $y = x[n] - x[n-1]$ |
| 3-pt moving average | $y = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$ |
| accumulator | $y[n] = y[n-1] + x[n]$ |

**Nonrecursive (FIR)**: inputs only. **Recursive (IIR)**: also past outputs. Needs a start value.
**Trap:** A $D$ that feeds $y$ back into the adder means $y[n-1]$ in the equation.

**WORK**

### *Steps — diagram → equation*

**1. NAME EACH WIRE**
After $k$ delays, a wire carries $x[n-k]$ or $y[n-k]$.

**2. EACH ADDER**
output = sum of the wires in, each times its gain.

**3. CHECK THE COUNT**
One $D$ per shift. One term per wire into the adder.

### *Steps — equation → diagram*

**1. ONE BLOCK PER PIECE**
Each $x[n-k]$: $k$ delays on the input side. Each $y[n-k]$: $k$ delays on the output side (feedback).

**2. GAINS**
One gain box per coefficient.

**3. ADDER**
All terms meet in one adder. Its output is $y[n]$.

**4. READ IT BACK**
Diagram → equation must give the original.

# 12 — system properties

**THE IDEA**
Properties are yes/no questions about a machine. Each is separate: a machine can pass one and fail another.

## Memory, causality, invertibility

**EXPLANATION**

### *Pieces: the time stamps*

| Stamp on $x[\cdot]$ | Color |
|---|---|
| $n$ (or $t$) | now |
| $n-1, n-2, \ldots$ | past |
| $n+1, \ldots,$ or $-n, 2n$ | future (for some $n$) |

**Trap:** $y(t) = x(-t)$: at $t = -1$ it needs $x(1)$, the future.
**Trap:** $y = x^2$ is not invertible: $1$ and $-1$ both give $1$.

**WORK**

**1. COLOR EACH STAMP**
now, past, or future.

↓

**2. MEMORYLESS?**
Every stamp is $\boxed{\text{now}}$ ⇒ memoryless. Else it has memory.

↓

**3. CAUSAL?**
No stamp is $\boxed{\text{future}}$ ⇒ causal.

↓

**4. INVERTIBLE?**
Solve for $x$. Exactly one answer ⇒ invertible. *(find two inputs with the same output to show no)*

## Linearity: mix before the machine, or after?

**EXPLANATION**

**Linear** means: mixing two inputs *before* the machine gives the same output as running each one alone and mixing the outputs *after*. Mixing means $c_1x_1 + c_2x_2$ for any numbers $c_1, c_2$.

### *Pieces*

| Piece | Where to find it |
|---|---|
| $\square$ | the input box: every place the recipe has an $x$. Keep the index on it: $\square[n-1]$ |
| $x_1, x_2$ | two test inputs. Never pick numbers: keep them as letters |
| $c_1, c_2$ | the knobs: any two numbers |
| $y_1, y_2$ | the recipe run on $x_1$ alone, and on $x_2$ alone |

### *Shortcut: look at how $x$ enters each term*

| Term in the recipe | Linear? |
|---|---|
| a number or a function of $n$ times one $x$: $3x[n]$, $nx[n]$, $\frac{x[n]}{n}$, $e^n x[n]$ | yes |
| $x$ shifted, flipped, or squeezed: $x[n-1]$, $x[-n]$, $x[2n]$ | yes |
| $x$ inside a function: $x^2$, $\sqrt{x}$, $e^x$, $\lvert x\rvert$, $\cos(x[n])$ | no |
| two $x$'s multiplied: $x[n]\,x[n-1]$ | no |
| a term with no $x$: $+3$, $+\sin n$, $+u[n]$ | no (fails the zero test) |

Every term yes ⇒ linear. One no ⇒ not linear. The steps on the right prove it.

| Quick example | Work → answer |
|---|---|
| $y = nx[n]$ | P1: $n(c_1x_1 + c_2x_2)$ P2: $c_1nx_1 + c_2nx_2$ equal |
| $y = x[n]^2$ | P1: $c_1^2x_1^2 + 2c_1c_2x_1x_2 + c_2^2x_2^2$ P2: $c_1x_1^2 + c_2x_2^2$ not equal |
| $y = 2x[n] + 3$ | zero test: $2(0) + 3 = 3 \neq 0$ not linear |

**Trap:** The coefficients may depend on $n$. Only how $x$ appears matters.
**Trap:** Use knobs $c_1, c_2$, not $a, b$. The system may have its own $a$.
**Trap:** Passing the zero test alone does not prove linear.

**WORK**

**1. BOX THE INPUT**
Rewrite the recipe with an empty box wherever it says $x$. Keep each index.
$y[n]$ = recipe of $\square$ for example $n \cdot \square[n] + \square[n-1]$

↓

**2. ZERO TEST**
Put $0$ in every box. If $y \neq 0$ ⇒ not linear. Stop.

↓

**3. PATH 1: MIX, THEN MACHINE**
Put $(\boxed{c_1}\boxed{x_1} + \boxed{c_2}\boxed{x_2})$ in every box, with that box's index.
Expand: multiply out every bracket. Call it **P1**.

↓

**4. PATH 2, PART A: MACHINE ON EACH**
$\boxed{y_1}$ = recipe with $x_1$ in every box.
$\boxed{y_2}$ = recipe with $x_2$ in every box.

↓

**5. PATH 2, PART B: THEN MIX**
P2 = $\boxed{c_1}\boxed{y_1} + \boxed{c_2}\boxed{y_2}$. Expand.

↓

**6. COMPARE TERM BY TERM**
Match each term of P1 with a term of P2.
All match, for every $c_1, c_2$ ⇒ linear.
A leftover term ($c_1^2$, $c_1c_2x_1x_2$, an extra constant) ⇒ not linear. The leftover is your reason.

## Time invariance: delay before the machine, or after?

**EXPLANATION**

**Time invariant** means: delaying the input by $n_0$ just delays the output by $n_0$. The machine ignores the clock.

### *Pieces*

| Piece | Where to find it |
|---|---|
| $\square$ | the input box: every place the recipe has an $x$ |
| inside of $x$ | the index in each box: $n$, $n-1$, $2n$, ... |
| lone $n$ | any $n$ (or $t$) outside a box: $nx[n]$, $\frac{x[n]}{n}$ |
| $n_0$ | the delay you apply |

### *Shortcut: look for the clock*

| Recipe has | Time invariant? |
|---|---|
| only shifts inside: $x[n-1]$, $x[n+2]$ | yes |
| a lone $n$: $nx[n]$, $\cos(n)\,x[n]$ | no |
| a squeeze or flip inside: $x[2n]$, $x[-n]$ | no |
| numbers only outside: $3x[n]$, $x^2$ | yes |

**Trap:** Path 1 changes only the inside of $x$. Path 2 changes every $n$.

**WORK**

**1. BOX THE INPUT**
Rewrite the recipe with $\square[\text{inside}]$ wherever it says $x$. Mark each lone $n$.

↓

**2. PATH 1: DELAY, THEN MACHINE**
In each box only: $\text{inside} \to \text{inside} - n_0$.
Leave every lone $n$ alone.

↓

**3. PATH 2: MACHINE, THEN DELAY**
Replace every $n$ with $n - n_0$: inside the boxes *and* each lone $n$.

↓

**4. COMPARE**
Same ⇒ time invariant. Different ⇒ time varying.

## Stability with a ceiling $B$

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $\lvert c_i \rvert$ | the size of each coefficient |
| $B$ | the input ceiling: $\lvert x[n] \rvert \le B$ |

**BIBO**: bounded in ⇒ bounded out.

| Red flag | Why |
|---|---|
| $nx[n]$ | $\lvert n \rvert B$ grows with $n$ |
| accumulator | adds another $B$ every step |
| $y = ay[n-1] + x$, $\lvert a \rvert \ge 1$ | the echo never shrinks |

**WORK**

**1. ABSOLUTE VALUE**
Take $\lvert\cdot\rvert$ of both sides.

↓

**2. SPLIT**
$\lvert a+b \rvert \le \lvert a \rvert + \lvert b \rvert$, $\lvert cx \rvert = \lvert c \rvert \lvert x \rvert$

↓

**3. CAP EACH INPUT**
$\lvert y \rvert \le \lvert c_1 \rvert B + \lvert c_2 \rvert B + \cdots$ *(shifted x too)*

↓

**4. VERDICT**
A finite number (no $n$, no endless sum) ⇒ stable.
To show unstable: one capped input ($x = 1$, $u[n]$) with an unbounded output.

# 13 — the convolution sum

**THE IDEA**

Every DT input is a stack of buttons. Press one button and an LTI machine answers with its echo, the **impulse response** $h[n]$. Each input stem makes its own echo, scaled by its height and shifted to its time. The output is the sum of all the echoes: $y = x * h$.

## Find $h$, and read the system from $h$

**EXPLANATION**

### *Pieces*

| Piece | Where to find it |
|---|---|
| $b_k$ | the coefficient of $x[n-k]$ |
| $k$ | the delay of that term |
| $x = \delta$ | the test input: $1$ at $n = 0$, else $0$ |

| From $h$ | Property |
|---|---|
| $h[n] = 0$ for $n < 0$ | causal |
| $\sum \lvert h[n] \rvert < \infty$ | BIBO stable |
| $h = c\,\delta[n]$ | memoryless |
| finitely many stems | FIR (nonrecursive) |
| series / parallel | $h_1 * h_2$ / $h_1 + h_2$ |

**Trap:** Only for LTI systems at rest. $y = x^2$ has no $h$.

**WORK**

### *Steps — nonrecursive → $h$*

**1. READ THE COEFFICIENTS**

$y = \sum \boxed{b_k}\, x[n - \boxed{k}]$

↓

**2. ONE STEM EACH**

$h[n] = \sum \boxed{b_k}\, \delta[n - \boxed{k}]$

### *Steps — recursive → $h$*

**1. PRESS THE BUTTON**

Put $\boxed{x = \delta}$. Start at rest: $h[-1] = 0$.

↓

**2. RUN IT**

$h[0]$, $h[1]$, $h[2]$, … one step at a time.

↓

**3. PATTERN**

Usually $h = a^n u[n]$.

### *Steps — $h$ → equation*

**1. ONE TERM PER STEM**

$y[n] = \sum_k h[k]\, x[n-k]$

## Convolve $x$ and $h$

EXPLANATION

### *Pieces*

| Piece | Where to find it |
|---|---|
| $x[k]$ | each input stem (Method 1: one color per stem) |
| $h$ | the echo, as stems or a formula |
| $n$ | the output time. Fixed during one sum |

| Fact | Formula |
|---|---|
| convolution sum | $y[n] = \sum_k x[k]\,h[n-k]$ |
| swap | $x * h = h * x$ |
| $x$ on $[a,b]$, $h$ on $[c,d]$ | $y$ on $[a+c,\, b+d]$ |
| length | $L_x + L_h - 1$ |
| sum check | $\sum y = \left(\sum x\right)\left(\sum h\right)$ |
| $x * \delta[n-n_0]$ | $x[n-n_0]$ |
| $a^n u * b^n u$, $a \neq b$ | $\dfrac{a^{n+1} - b^{n+1}}{a-b}\,u[n]$ |
| $a^n u * a^n u$ | $(n+1)a^n u[n]$ |
| finite geometric | $\sum_{k=0}^{n} r^k = \dfrac{1-r^{n+1}}{1-r}$ |

| If the signals are | Use |
|---|---|
| two short lists of stems | Method 1, table |
| short pulse, long formula | Method 2, flip and slide |
| two formulas with switches | Method 3, formula |

**Trap:** $h[n-k]$, not $h[k-n]$. And $y \neq x \cdot h$.
**Trap:** After the flip, $h[0]$ sits at $k = n$, and $h[1]$ at $k = n-1$.

WORK

### *Method 1 — the table*

**1. RANGE AND LENGTH**
$y$ on $[a+c,\, b+d]$, length $L_x + L_h - 1$.

↓

**2. ONE ROW PER INPUT STEM**
row $k$: $\boxed{x[k]} \cdot \boxed{h}$, starting at column $n = k$

↓

**3. ADD EACH COLUMN**
$y[n]$ = sum of column $n$

↓

**4. CHECK**
$\sum y = \left(\sum x\right)\left(\sum h\right)$

### *Method 2 — flip and slide*

**1. DRAW $x[k]$**
on a $k$ axis

↓

**2. FLIP AND SLIDE $h$**
$h[k] \to h[-k] \to h[\boxed{n} - k]$. Its first stem sits at $k = n$, the rest trail left.

↓

**3. REGIONS**
Split $n$: no overlap ($y = 0$), partial overlap, full overlap.

↓

**4. EACH REGION**
$y[\boxed{n}] = \sum_{k \in \text{overlap}} \boxed{x[k]}\,\boxed{h[n-k]}$

↓

**5. CHECK**
The region formulas agree where they meet.

### *Method 3 — formula*

**1. PLACE**
$y[n] = \sum_k \boxed{x[k]}\,\boxed{h[n-k]}$

↓

**2. SWITCHES CUT THE SUM**
$u[k]\,u[n-k] \Rightarrow k = 0..\boxed{n}$, and $y = 0$ for $n < 0$.

↓

**3. PULL OUT, GEOMETRIC SUM**
Pull out what has no $k$. Then $\sum_{k=0}^{n} r^k = \dfrac{1-r^{n+1}}{1-r}$.

↓

**4. CHECK**
$y[0] = x[0]\,h[0]$.